

CUSTOMER PROFILING, ANOMALY DETECTION AND COST BENEFIT ANALYTICS FOR SOLAR ROOFTOP ADOPTION IN POWER DISCOMS

Consumer Profiling, Load Anomaly Detection, PCA, CBLOF and Rooftop Solar Suitability

Hansot Subdivision – Machine Learning & Data Analysis Project

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1. Abstract

This project develops an electricity-consumption intelligence workflow using consumer-level data from Hansot subdivision. The project combines data analysis with course-aligned machine-learning methods: KNN classification, Decision Tree classification, K-Means clustering, PCA and CBLOF-style anomaly detection. The supplied working dataset contains 13,650 consumer records and ten monthly consumption columns covering November 2025 to August 2026. A future-period classification proxy uses the first seven months to predict whether the final three-month average reaches a high-consumption threshold. Hyperparameter tuning is used to improve KNN and Decision Tree performance. K-Means and PCA support consumer profiling and visualization, while CBLOF provides a clustering-based anomaly score. A transparent load-ratio benchmark is retained for comparison. Rooftop solar analysis is presented as preliminary consumption-based scenario analysis rather than certified technical feasibility.

2. Problem Statement

Large consumer-level electricity datasets contain useful patterns but can be difficult to inspect manually. The project converts monthly consumption and contract-load information into interpretable features and ML outputs for consumer classification, behavioral segmentation, anomaly screening and preliminary solar suitability.

3. Course Alignment

The supplied syllabus explicitly covers distance measures, K-Means and other clustering methods, anomaly detection/CBLOF, KNN classification, Decision Trees, hyperparameter tuning, PCA, neural networks and ensemble learning. The implemented core therefore uses KNN, Decision Tree, K-Means, PCA and CBLOF-style anomaly detection.

4. Dataset

The supplied workbook contains 13,650 records, ten monthly consumption columns from November 2025 through August 2026, contract load, tariff, area, feeder, village and solar-consumer status. It contains 20 feeders, 50 villages, 50 AREA values, 562 solar consumers, 13,088 non-solar consumers and 2,306 missing monthly cells.

5. Data Preparation and Feature Engineering

The project creates first-seven-month average, standard deviation, minimum, maximum, total, coefficient of variation and consumption-per-contract-load features. Future-period consumption is used only to construct the classification target. Because billing dates and billing-period metadata are not present, the project does not automatically reinterpret alternate-month zeros as bimonthly bills.

6. Classification Target

The first seven months are treated as the historical predictor period and the final three months as the future evaluation period. The high-consumption threshold is the 75th percentile of first-seven-month average consumption. This is a project-defined forecasting proxy and should not be described as a validated new-connection prediction model.

7. KNN Classification and Decision Tree

KNN uses standardized numeric features, one-hot encoded categorical features and grid-search tuning over k, weights and distance parameter p. The Decision Tree uses grid-search tuning over maximum depth and minimum samples per leaf. Both use the same stratified 75/25 held-out test set.

8. Tuned Classification Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
KNN (tuned)					
	0.8400				
		0.8166			
			0.7587		
				0.7866	
					0.9080
Decision Tree (tuned)					
	0.9045				
		0.8587			

			0.9027		
				0.8801	
					0.9651
Model	Accuracy	Precision	Recall	F1	ROC-AUC
KNN (tuned)	0.840	0.817	0.759	0.787	0.908
Decision Tree (tuned)	0.904	0.859	0.903	0.880	0.965

The reported results are tied to the stated target, feature set, split and tuned search space. They are not universal estimates of model performance.

9. Hyperparameter Tuning

Hyperparameter experiments are stored in results/hyperparameter_experiments.csv. KNN tuning varies the number of neighbors, distance weighting and Minkowski distance parameter. Decision Tree tuning varies maximum depth and minimum samples per leaf. The selected configuration is the one with the highest cross-validated F1 score within the specified search space.

10. K-Means Clustering

K-Means is applied to standardized behavioral/load features. Candidate K values are compared using silhouette score. The selected K is the highest-scoring candidate among the evaluated values. Cluster interpretation must also consider cluster sizes and feature profiles.

Evaluated clustering results:

K	Silhouette
2	0.991
3	0.529
4	0.455
5	0.457
6	0.464

11. PCA

PCA is applied after standardizing six behavioral/load features. The first two components are used to visualize consumer profiles and K-Means structure. PCA is a dimensionality-reduction and visualization tool here; it is not itself a prediction model.

PC1 explains 49.97% of standardized-feature variance and PC2 explains 18.34%.

12. CBLOF Anomaly Detection

CBLOF is implemented using K-Means cluster structure. Large clusters are identified using alpha/beta size rules, and each consumer receives a score based on distance to the nearest large cluster. The top 1% of CBLOF scores are flagged as a configurable review set. Because verified anomaly labels are unavailable, the CBLOF result cannot be expressed as detection accuracy. It should be validated by expert review.

The development run flagged 137 records at the 99th-percentile CBLOF threshold.

13. Load-Ratio Baseline

The transparent baseline uses: $\text{Benchmark Energy} = \text{Contract Load (kW)} \times 8 \text{ hours/day} \times \text{billing days} \times 0.8$. $\text{Consumption Ratio} = \text{actual energy} / \text{benchmark}$. This is an assumption-based screening reference, not a physical maximum/minimum and not proof of theft or unauthorized load.

14. Rooftop Solar Suitability

The dashboard includes a preliminary solar scenario calculator for annual generation, on-site use, export, gross/net cost, savings and simple payback. The dataset lacks rooftop area, shading, daytime load share, tariff rates and export compensation, so technical feasibility requires additional site information.

15. Limitations

No billing-period metadata; no verified anomaly labels; no connection dates/post-connection history; no measured peak demand or contract demand in kVA; no rooftop area/shading/daytime-load information; and the classification target is a project-defined future-period proxy.

16. Individual Contributions

Replace placeholders with the team's actual names and actual work performed before submission.

Member	Area	Specific contribution
Bony Thomas, 25280041	Data & EDA	Data validation, cleaning, feature engineering, exploratory analysis and consumer/geographic profiling.
Bhavin Kirtikumar Patel–25280040	ML Models	KNN, Decision Tree, hyperparameter tuning, K-Means, PCA and model evaluation.
Mayurkumar Bholabhai Bhalani-25280045	Anomaly/Solar/Dashboard	CBLOF, load-ratio anomaly baseline, solar calculations, Streamlit dashboard, integration and GitHub documentation.

17. Future Scope

Add verified anomaly investigation labels, connection dates and post-connection history, interval/AMI data, measured peak demand, rooftop technical information, and additional syllabus-approved models such as neural networks or ensemble methods where justified. CBLOF thresholds should be calibrated using field validation.

18. Conclusion

The revised project now maps directly to the supplied syllabus: KNN and Decision Tree provide supervised classification, K-Means provides clustering, PCA supports dimensionality reduction and visualization, and CBLOF provides syllabus-aligned anomaly detection. Hyperparameter tuning makes model improvement measurable and reproducible. The system is packaged as a GitHub-ready repository while explicitly separating data-supported findings from assumptions.